



Figure 6: Percentage of each head’s attention focused on amino acids *Pro* (left) and *Phe* (right).



Figure 7: Pairwise attention similarity (left) vs. substitution matrix (right) (codes in App. C.5)

is encoded in attention weights primarily in the last 1-2 layers. These results are consistent with prior work in NLP that suggests deeper layers in text-based Transformers attend to more complex properties (Vig & Belinkov, 2019) and encode higher-level representations (Raganato & Tiedemann, 2018; Peters et al., 2018; Tenney et al., 2019; Jawahar et al., 2019).

The *embedding* probes (Figure 5, orange) also show that the model first builds representations of local secondary structure in lower layers before fully encoding binding sites and contact maps in deeper layers. However, this analysis also reveals stark differences in how knowledge of contact maps is accrued in embeddings, which accumulate this knowledge gradually over many layers, compared to attention weights, which acquire this knowledge only in the final layers in this case. This example points out limitations of common layerwise probing approaches that only consider embeddings, which, intuitively, represent what the model *knows* but not necessarily how it operationalizes that knowledge.

4.4 AMINO ACIDS AND THE SUBSTITUTION MATRIX

In addition to high-level structural and functional properties, we also performed a fine-grained analysis of the interaction between attention and particular amino acids.

Attention heads specialize in particular amino acids. We computed the proportion of TapeBert’s attention to each of the 20 standard amino acids, as shown in Figure 6 for two example amino acids. For 16 of the amino acids, there exists an attention head that focuses over 25% of attention on that amino acid, significantly greater than the background frequencies of the corresponding amino acids, which range from 1.3% to 9.4%. Similar behavior was observed for ProtBert, ProtBert-BFD, ProtAlbert, and ProtXLNet models, with 17, 15, 16, and 18 amino acids, respectively, receiving greater than 25% of the attention from at least one attention head. Detailed results for TapeBert including statistical significance tests and comparison to a null model are presented in Appendix C.5.

Attention is consistent with substitution relationships. A natural follow-up question from the above analysis is whether each head has “memorized” specific amino acids to target, or whether it has actually learned meaningful properties that correlate with particular amino acids. To test the latter hypothesis, we analyze whether amino acids with similar structural and functional properties are attended to similarly across heads. Specifically, we compute the Pearson correlation between the distribution of attention across heads between all pairs of distinct amino acids, as shown in Figure 7 (left) for TapeBert. For example, the entry for *Pro* (P) and *Phe* (F) is the correlation between the two heatmaps in Figure 6. We compare these scores to the BLOSUM62 substitution scores (Sec. 2) in Figure 7 (right), and find a Pearson correlation of 0.73, suggesting that attention is moderately

consistent with substitution relationships. Similar correlations are observed for the ProtTrans models: 0.68 (ProtBert), 0.75 (ProtBert-BFD), 0.60 (ProtAlberty), and 0.71 (ProtXLNet). As a baseline, the randomized versions of these models (Appendix B.2) yielded correlations of -0.02 (TapeBert), 0.02 (ProtBert), -0.03 (ProtBert-BFD), -0.05 (ProtAlberty), and 0.21 (ProtXLNet).

5 RELATED WORK

5.1 PROTEIN LANGUAGE MODELS

Deep neural networks for protein language modeling have received broad interest. Early work applied the Skip-gram model (Mikolov et al., 2013) to construct continuous embeddings from protein sequences (Asgari & Mofrad, 2015). Sequence-only language models have since been trained through autoregressive or autoencoding self-supervision objectives for discriminative and generative tasks, for example, using LSTMs or Transformer-based architectures (Alley et al., 2019; Bepler & Berger, 2019; Rao et al., 2019; Rives et al., 2019). TAPE created a benchmark of five tasks to assess protein sequence models, and ProtTrans also released several large-scale pretrained protein Transformer models (Elnaggar et al., 2020). Riesselman et al. (2019); Madani et al. (2020) trained autoregressive generative models to predict the functional effect of mutations and generate natural-like proteins.

From an interpretability perspective, Rives et al. (2019) showed that the output embeddings from a pretrained Transformer can recapitulate structural and functional properties of proteins through learned linear transformations. Various works have analyzed output embeddings of protein models through dimensionality reduction techniques such as PCA or t-SNE (Elnaggar et al., 2020; Biswas et al., 2020). In our work, we take an interpretability-first perspective to focus on the internal model representations, specifically attention and intermediate hidden states, across multiple protein language models. We also explore novel biological properties including binding sites and post-translational modifications.

5.2 INTERPRETING MODELS IN NLP

The rise of deep neural networks in ML has also led to much work on interpreting these so-called black-box models. This section reviews the NLP interpretability literature on the Transformer model, which is directly comparable to our work on interpreting Transformer models of protein sequences.

Interpreting Transformers. The Transformer is a neural architecture that uses attention to accelerate learning (Vaswani et al., 2017). In NLP, transformers are the backbone of state-of-the-art pre-trained language models such as BERT (Devlin et al., 2019). *BERTology* focuses on interpreting what the BERT model learns about language using a suite of probes and interventions (Rogers et al., 2020). So-called *diagnostic classifiers* are used to interpret the outputs from BERT’s layers (Veldhoen et al., 2016). At a high level, mechanisms for interpreting BERT can be placed into three main categories: interpreting the learned embeddings (Ethayarajh, 2019; Wiedemann et al., 2019; Mickus et al., 2020; Adi et al., 2016; Conneau et al., 2018), BERT’s learned knowledge of syntax (Lin et al., 2019; Liu et al., 2019; Tenney et al., 2019; Htut et al., 2019; Hewitt & Manning, 2019; Goldberg, 2019), and BERT’s learned knowledge of semantics (Tenney et al., 2019; Ettinger, 2020).

Interpreting attention specifically. Interpreting attention on textual sequences is a well-established area of research (Wiegrefe & Pinter, 2019; Zhong et al., 2019; Brunner et al., 2020; Hewitt & Manning, 2019). Past work has been shown that attention correlates with syntactic and semantic relationships in natural language in some cases (Clark et al., 2019; Vig & Belinkov, 2019; Htut et al., 2019). Depending on the task and model architecture, attention may have less or more explanatory power for model predictions (Jain & Wallace, 2019; Serrano & Smith, 2019; Pruthi et al., 2020; Moradi et al., 2019; Vashishth et al., 2019). Visualization techniques have been used to convey the structure and properties of attention in Transformers (Vaswani et al., 2017; Kovaleva et al., 2019; Hoover et al., 2020; Vig, 2019). Recent work has begun to analyze attention in Transformer models outside of the domain of natural language (Schwaller et al., 2020; Payne et al., 2020).

Our work extends these methods to protein sequence models by considering particular biophysical properties and relationships. We also present a joint cross-layer probing analysis of attention weights and layer embeddings. While past work in NLP has analyzed attention and embeddings across layers, we believe we are the first to do so in any domain using a single, unified metric, which enables us to

directly compare the relative information content of the two representations. Finally, we present a novel tool for visualizing attention embedded in three-dimensional structure.

6 CONCLUSIONS AND FUTURE WORK

This paper builds on the synergy between NLP and computational biology by adapting and extending NLP interpretability methods to protein sequence modeling. We show how a Transformer language model recovers structural and functional properties of proteins and integrates this knowledge directly into its attention mechanism. While this paper focuses on reconciling attention with known properties of proteins, one might also leverage attention to uncover novel relationships or more nuanced forms of existing measures such as contact maps, as discussed in Section 4.1. In this way, language models have the potential to serve as tools for scientific discovery. But in order for learned representations to be accessible to domain experts, they must be presented in an appropriate context to facilitate discovery. Visualizing attention in the context of protein structure (Figure 1) is one attempt to do so. We believe there is the potential to develop such contextual visualizations of learned representations in a range of scientific domains.

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